

## Tut 8

3. Consider a system with the following characteristics.

- Direct mapped cache of 32 cache blocks and cache block size of 32 bytes
  - Cache uses Physical Address for address mapping
  - Virtual Memory page size 2048 bytes
  - Virtual Memory size is 1Mbyte. Physical Memory size is 64KByte
  - Extracts of Page Table (valid entries)
    - Virtual Page 0 → Physical Frame 9
    - Virtual Page 1 → Physical Frame 3
    - Virtual Page 2 → Physical Frame 5
    - Virtual Page 3 → Physical Frame 2
    - Virtual Page 4 → Physical Frame 7
  - The main program is 5KByte in size and starts at virtual address 0x01006
- (i) Assuming that the compiler allocates the program sequentially in the virtual memory, what is the physical address of the start and end of the main program?

Virtual address (20bits): start=0x01006 = 0001 0 **000 0000 0110** (11 bits for virtual memory page size of 2048) → Page 2 → Physical Frame 5 (101b) → physical address= 10 1 **000 0000 0110** = 0x2806 (Result 1)

The length of 5KB in binary representation: 101 00 0000 0000=0x1400;  
**end=0x2806 (Result 1) + 0x1400 - 1 = 0x3C05 (Result)**

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Continued from the  
previous slide ...

**Physical address = 0010 1000 0000 0110 = 0x2806**

Start address...from the  
previous slide ...

- (ii) Which cache block should the CPU check in the cache for the start of the main program? What is the corresponding TAG value used to check for cache hit/miss?

**Physical Address used for tagging. Hence, TAG:BLK:OFFSET = 6:5:5**

**Start address = 0x2806 = 0010 1000 0000 0110**

**TAG value = 001010 = 0xA, BLK = 000000 = 0**

## Tut 9

4. You have been tasked to write a program that calculates the actual time based on a counter that is incremented once every 0.10 seconds. For example, if the counter value is 3,600,000, you would expect the actual time to be 100 hours  $((3,600,000 \times 0.1) / (60 \times 60))$ .

Suppose you have decided to use a 24-bit fixed point representation as shown in Figure 8.4 to store the value of 0.10 seconds  $(2^{-4} + 2^{-5} + 2^{-8} + 2^{-9} + 2^{-12} + 2^{-13} + \dots)$ .

|   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | ▪ | 0 | 0 | 0 | 1 | 1 | 0 | 0 | 1 | 1 | 0 | 0 | 1 | 1 | 0 | 0 | 1 | 1 | 0 | 0 | 1 | 1 | 0 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

**Figure 8.4 – Fixed point representation of 0.10**

- Approximate the round-off error (in decimal) of  $0.10_{10}$  due to the fixed-point representation.
- What is the effect of this round-off error on the time calculated if the counter value is 3,600,000?

|                         |   |   |   |   |          |          |   |   |          |          |   |   |           |           |   |   |           |           |   |   |           |           |   |   |
|-------------------------|---|---|---|---|----------|----------|---|---|----------|----------|---|---|-----------|-----------|---|---|-----------|-----------|---|---|-----------|-----------|---|---|
| 0                       | ▪ | 0 | 0 | 0 | 1        | 1        | 0 | 0 | 1        | 1        | 0 | 0 | 1         | 1         | 0 | 0 | 1         | 1         | 0 | 0 | 1         | 1         | 0 | 0 |
|                         |   |   |   |   | $2^{-4}$ | $2^{-5}$ |   |   | $2^{-8}$ | $2^{-9}$ |   |   | $2^{-12}$ | $2^{-13}$ |   |   | $2^{-16}$ | $2^{-17}$ |   |   | $2^{-19}$ | $2^{-20}$ |   |   |
| (Given by the question) |   |   |   |   |          |          |   |   |          |          |   |   |           |           |   |   |           |           |   |   |           |           |   |   |

Approximation for  $0.1_{10} = 0.09999999046325684_{10}$

Round-off Error =  $0.10_{10} - 0.09999999046325684_{10} = 0.0000000095_{10}$

For counter value of 3600000, the time calculated will be off by  
 $3600000_{10} \times 0.0000000095_{10} = \mathbf{0.342 \text{ seconds}}$

Missile speed = 1676 meter/second  
 0.342 seconds translate to  $0.342 \times 1676 = 573 \text{ meter}$